Scientific Evaluation — ERC Starting Grant 2027

SpecTrust: Spectral Duality for Trustworthy Perception and Generation
PI: Prof. Margret Keuper | Host: University of Mannheim | Duration: 60 months
Independent simulated ERC panel-style scientific assessment of Parts B1 and B2; not an official ERC decision.

Executive assessment and funding recommendation

Recommendation: Borderline / reserve rather than immediate funding in its present form. SpecTrust puts forward an original, high-risk/high-gain hypothesis linking discriminative robustness and generative faithfulness through recoverable task information and uncertainty. It combines excellent PI expertise with a coherent research programme. The central concern is whether the proposed measurements and experiments can conclusively test the claimed unifying mechanism.

Criterion	Indicative assessment
Originality and ambition	9.0 / 10 — very strong
Breakthrough potential	8.5 / 10 — high
Conceptual clarity	8.0 / 10 — strong
Methodological rigor	6.5 / 10 — mixed
Feasibility and focus	7.5 / 10 — moderate to strong
PI expertise and leadership	9.5 / 10 — outstanding

Indicative scores are diagnostic judgments only; ERC does not use this numerical scorecard.

Principal strengths

S1. Original and potentially unifying hypothesis — major strength

A compelling parallel is drawn between discriminative systems that remain confident after relevant evidence degrades and generative systems that produce unsupported detail. Separating competence (R) from warranted confidence (CΦ) avoids equating good calibration with useful task performance. If validated, this could provide a genuinely general explanatory framework.

S2. Controlled discriminative–generative comparison — major strength

WP2a compares depth, flow, disparity, and segmentation models against shared ground truth, using common output representations and spatial or frequency-resolved errors. This is substantially stronger than comparing unrelated classification and generation benchmarks.

S3. Coherent sequence from measurement to intervention

WP1 develops and validates measurements; WP2 tests generalization across classes; WP3 uses the findings to guide architectures and training. Comparison with uniform, permuted, and inverted filter controls is valuable, even if it does not fully establish causation.

S4. Outstanding PI fit and research foundation

The selected publications and track record directly cover frequency-domain vision, aliasing and robustness, uncertainty, representation collapse, generative modelling, dense prediction, and semantic representations. Leadership and supervision experience support implementation feasibility.

S5. Falsifiable predictions and explicit risk awareness

The application states three predictions, includes benchmark comparisons and two go/no-go decision points, and recognizes the prospect of negative or class-specific findings. This is well suited to exploratory ERC research.

Principal scientific weaknesses — ranked by importance

W1. CΦ does not directly operationalize the full central hypothesis — critical

The key score correlates uncertainty u_f(p) with a probe's reconstruction error e_f(p) within a band. Probe error can reflect decoder capacity or training rather than absent information; within-band spatial alignment differs from the claimed across-band recoverability–uncertainty relation; and rank correlation is invariant to monotone miscalibration. Positive results might therefore establish a useful error-ranking indicator rather than the proposed explanatory principle.

Action: Define and test separately (i) information recoverability, (ii) spatial/semantic uncertainty–error alignment, and (iii) the calibration of uncertainty magnitudes. State which combination constitutes decisive evidence for the hypothesis.

W2. Jacobian sensitivity is not equivalent to recoverable information — critical

Task-conditioned Jacobian norms describe local differential sensitivity, not global decodability or mutual information. Low sensitivity can indicate nuisance invariance, while high sensitivity need not imply useful information. Normalizing spectral sensitivity across bands can also obscure global changes. Probe–Jacobian disagreement will not have a unique interpretation.

Action: Present the Jacobian as a complementary diagnostic, justify any stronger recoverability interpretation, and validate both estimates on synthetic or controlled transformations with known information loss.

W3. Uncertainty is not directly comparable across model classes — critical

Discriminative ensemble variation and generative sample diversity need not encode the same kind of uncertainty. Generative variance also depends on sampler settings, legitimate multimodality, and possible mode collapse. These confounds matter to the proposed cross-class equivalence.

Action: Validate uncertainty estimators against held-out error and distinguish epistemic uncertainty, task ambiguity, and sampling variability before making cross-class conclusions.

W4. WP3 does not isolate CΦ as the causal mediator — major

Improved filtering may help via anti-aliasing, regularization, optimization, or feature selection rather than through restoration of the recoverability–uncertainty match. Gains are computed from R and U, not directly from CΦ; improvement in both CΦ and robustness remains observational at the mediator level.

Action: Compare interventions that alter R, uncertainty alignment, and calibration separately; test whether changes in CΦ explain gains beyond changes in ordinary performance and spectral stability.

W5. WP2b semantic extension is less controlled — major

Reference classifiers, attribute questions, sparse autoencoders, and semantic hierarchies introduce additional measurement assumptions. For underspecified text prompts, variable generated details may be legitimate rather than unfaithful. SAE feature hierarchies need not correspond neatly to WordNet levels.

Action: Use explicit prompt constraints and validated semantic targets for primary inference; position exploratory SAE analyses as extensions rather than decisive evidence of the general principle.

W6. Statistical inference is underspecified — major

Approximately ten public models per task and class, plus related checkpoints, may yield limited independent model-level evidence for cross-model correlations, nested regressions, and equivalence margins. Numerous images are not substitutes for independent models.

Action: Prespecify effect sizes, equivalence margins, hierarchical or clustered analyses, uncertainty intervals, and model-family sensitivity checks.

W7. Breadth stretches team capacity — moderate

The programme spans multiple dense prediction tasks, two model families, Jacobians, calibration, VLMs, text-to-image models, MSAEs, and intervention design. This is ambitious for two PhD researchers, one postdoc, and 50% PI effort.

Action: Prioritize the essential WP1–WP2a–WP3a pathway, with WP2b and WP3b conditional on validated core measurements.

Work-package assessment

WP1 — Recoverability and uncertainty

Highest original methodological contribution; also the project's critical dependency. Strong problem choice, but measurement validity must be established.

WP2a — Controlled cross-class testing

Strongest empirical component. Excellent shared-task setup; results depend on uncertainty-measure validation.

WP2b — Semantic and multimodal extension

Important potential reach, but materially less controlled and vulnerable to surrogate-evaluator and semantic-definition assumptions.

WP3a — Spectral architecture/training intervention

Feasible and well matched to PI expertise, with practical potential. Needs a more discriminating causal design.

WP3b — Semantic interventions

Interesting but exploratory and reliant on WP2b identifying meaningful and stable representations.

PI and scientific impact

PI: Outstanding fit for the research agenda, supported by field-relevant publications, sustained research leadership, international collaboration, and supervision experience. The principal limiting factor is not PI competence but the decisiveness of the proposed tests.

High-gain outcome: A replicable general principle predicts failures in both model classes and supports effective interventions — potentially field-changing. Intermediate outcome: Useful class- or task-specific measures, benchmarks, and spectral interventions — substantial research contribution. Unfavourable outcome: Inconsistent recoverability and uncertainty estimators or no gain beyond existing measures — the central unifying claim would remain unsupported.

Decision and priority revisions

Panel-style recommendation: Competitive but reserve / borderline in current form. A high-quality Step 1 outcome is plausible, while a full scientific evaluation would likely raise significant reservations about whether the methodology genuinely tests the central hypothesis. This is an illustrative prediction, not an official funding probability or panel grade.

1. Define the central match precisely

Separate across-band recoverability effects, within-band uncertainty/error alignment, and quantitative confidence calibration; identify disconfirming evidence.

2. Validate probe and Jacobian interpretations

Explicitly distinguish recoverability from local sensitivity and use controlled information-loss tests.

3. Make WP2a the decisive experiment

Validate uncertainty comparability and provide robust model-level statistical inference; position WP2b as extension.

4. Strengthen WP3 causal isolation

Separate benefits of filtering or representation quality from those specifically mediated by the match.

Overall: Targeted revision, not a wholesale redesign, is recommended. The project already has ERC-level conceptual ambition and PI expertise. A more exact link between hypothesis, operational measurement, and decisive experiments could move it from borderline to a strong funding recommendation.

Basis of review: SpecTrust ERC Starting Grant 2027, Part B1 (15 pp., including cover, references, CV/track record) and Part B2 (15 pp., including references and Funding ID).